

SESSION 19

GitHub Deployment & PythonAnywhere Hosting

Topics Covered: Push Django project to GitHub, PythonAnywhere deployment, Static & media files config, Database migration on server

Activities: Fully deploy Doctor Finder project, Test live routes

Task 1: Create new GitHub repo named 'MyPlaylistApp', initialize local Django project, push full code to repository.

Explanation & Approach: *Initializes git tracking, sets up .gitignore, commits files, pushes to remote GitHub repository.*

```
# Terminal Commands:
git init
git add .
git commit -m "Initial commit - MyPlaylistApp Django Project"
git branch -M main
git remote add origin https://github.com/yourusername/MyPlaylistApp.git
git push -u origin main
```

Task 2: Deploy Django 'MyPlaylistApp' to PythonAnywhere, set up web app, ensure home page accessible via public URL.

Explanation & Approach: *Configures PythonAnywhere virtualenv, WSGI file, and web domain.*

```
# PythonAnywhere Deployment Walkthrough

1. Open PythonAnywhere Bash Console:
$ git clone https://github.com/yourusername/MyPlaylistApp.git
$ cd MyPlaylistApp
$ mkvirtualenv --python=/usr/bin/python3.10 myenv
$ pip install django

2. Configure Web Tab:
- Source code path: /home/yourusername/MyPlaylistApp
- Virtualenv path: /home/yourusername/.virtualenvs/myenv

3. Update WSGI Configuration File (/var/www/yourusername_pythonanywhere_com_wsgi.py):
import os
import sys
path = '/home/yourusername/MyPlaylistApp'
if path not in sys.path:
    sys.path.append(path)
os.environ['DJANGO_SETTINGS_MODULE'] = 'foodiehub_project.settings'
from django.core.wsgi import get_wsgi_application
application = get_wsgi_application()
```

Task 3: Configure static and media files on PythonAnywhere so uploaded images display correctly on live site.

Explanation & Approach: *Collects static files into STATIC_ROOT and configures web server mappings.*

```
# settings.py
STATIC_URL = '/static/'
STATIC_ROOT = os.path.join(BASE_DIR, 'staticfiles')

MEDIA_URL = '/media/'
MEDIA_ROOT = os.path.join(BASE_DIR, 'media')

# Run in PythonAnywhere console:
python manage.py collectstatic --noinput

# In PythonAnywhere Static Files Section:
# URL: /static/ -> Directory: /home/yourusername/MyPlaylistApp/staticfiles
# URL: /media/ -> Directory: /home/yourusername/MyPlaylistApp/media
```

Task 4: Run makemigrations and migrate commands on PythonAnywhere to create database tables, verify via Django admin.

Explanation & Approach: *Executes migrations against PythonAnywhere SQLite or MySQL production instance.*

```
# In PythonAnywhere Bash Console:
python manage.py makemigrations
python manage.py migrate
python manage.py createsuperuser
```

Task 5: Test at least two live routes of deployed app and document errors/issues encountered with resolution steps.

Explanation & Approach: *Documents deployment troubleshooting steps (ALLOWED_HOSTS, DEBUG settings).*

```
# Live Deployment Audit Log

Tested Routes:
1. https://yourusername.pythonanywhere.com/music/ -> HTTP 200 OK
2. https://yourusername.pythonanywhere.com/admin/ -> HTTP 200 OK

Troubleshooting Identified:
- Error: Invalid HTTP_HOST header: 'yourusername.pythonanywhere.com'.
- Resolution: Updated settings.py ALLOWED_HOSTS = ['yourusername.pythonanywhere.com', 'localhost']
- Reloaded Web app from PythonAnywhere dashboard.
```